C-STGB Neural Tensor Flow: Message Passing, Harmonic Encoders, Edge Gates & Fusion

STEP 1: ENCODER
3-Band Temporal

Raw Input Tensors

$$\mathbf{x}_u \in \mathbb{R}^d,\ \mathbf{e}_{uv} \in \mathbb{R}^{d_e}$$

Harmonic Band 1

$$\Phi_{\text{burst}}(\Delta t) \in \mathbb{R}^{d/3}$$

Harmonic Band 2

$$\Phi_{\text{diurnal}}(\Delta t)\ (24\text{h})$$

Harmonic Band 3

$$\Phi_{\text{macro}}(\Delta t)\ (90\text{d})$$

Hawkes State

$$\lambda_u(t) = \mu_0 + \sum \alpha e^{-\beta \Delta t}$$

$$\mathbf{h}_u^{(0)} \in \mathbb{R}^{d_{\text{embed}}}$$

STEP 2: EDGE GATE
Edge-Trust MLP

Edge Context Cat

$$[\mathbf{h}_u \parallel \mathbf{h}_v \parallel \mathbf{e}_{uv} \parallel \Phi_{\text{time}}]$$

Trust Scoring MLP

$$g_{ij} = \sigma(\mathbf{W}_g^\top [\cdot] + b_g)$$

Floor Regularizer

$$\hat{g}_{ij} = \delta_0 + (1 - \delta_0) g_{ij}$$

Threshold Floor

$$\delta_{\text{floor}} = 0.10 \text{ (Prunes 65.9\%)}$$

Denoised Graph

$$\tilde{\mathcal{E}} = \{(u, v) : \hat{g}_{uv} \geq 0.10\}$$

$$\mathbf{A}_{\text{gated}} = \hat{\mathbf{G}} \odot \mathbf{A}$$

STEP 3: MESSAGE PASS
Continuous HGT

Hetero Message

$$\mathbf{m}_{vu} = \mathbf{W}_v \mathbf{h}_v^{(l-1)} \hat{g}_{vu}$$

Harmonic Attention

$$\alpha_{vu} = \text{Softmax}(\mathbf{q}_u^\top \mathbf{k}_v / \sqrt{d})$$

Brody Curvature

$$\kappa(u, v) \text{ Riemannian bound}$$

Neighborhood Agg

$$\tilde{\mathbf{h}}_u = \sum_{v \in \mathcal{N}_u} \alpha_{vu} \mathbf{m}_{vu}$$

Feed-Forward Layer

$$\mathbf{h}_u^{(l)} = \text{LayerNorm}(\mathbf{h}_u + \text{MLP})$$

$$\mathbf{h}_u^{(L)} \in \mathbb{R}^d \text{ (Structural)}$$

STEP 4: GRAPH-SMOTE
Latent Typology

Typology Clusters

$$C_1, \ \ldots, C_K \ (\tau_{\text{sim}} \geq 0.75)$$

Intra-Cluster Interp

$$\mathbf{h}_{\text{syn}} = (1 - \rho)\mathbf{h}_u + \rho \mathbf{h}_v$$

Bilinear Link Head

$$\hat{A}_{\text{syn},\, w} = \sigma(\mathbf{h}_{\text{syn}}^\top \mathbf{W}_e \mathbf{h}_w)$$

Flow Invariant Check

$$\Phi_{\text{flow}}(\text{syn}) \approx 1.0 \text{ (Pass-thru)}$$

Train Isolation

100% Quarantined to D_{train}

$$\tilde{\mathcal{G}}_{\text{train}} \ (+9.92 \text{ pp Lift})$$

STEP 5: FUSION & HEAD
Cold-Start & Bayes

Universal Invariant

$$\mathbf{z}_{\text{inv}} \in \mathbb{R}^{12} \text{ (Kirchhoff)}$$

Evidence Gating

$$\alpha_u = \sigma(\mathbf{w}_\alpha^\top [\mathbf{h}_u \parallel \mathbf{z}_{\text{inv}}])$$

Fused Embedding

$$\mathbf{h}_u^* = \alpha_u \mathbf{h}_u + (1 - \alpha_u) \mathbf{z}_{\text{inv}}$$

Bayes Loss Policy

$$\mathcal{L}_{\text{asym}} = 15 \cdot \text{FN} + 1 \cdot \text{FP}$$

Optimal Cutoff

$$\tau^* = 0.18 \rightarrow \hat{y} = \mathbf{1}(\hat{p} \geq 0.18)$$

$$\hat{p}_u \in [0, 1] \rightarrow \Gamma(u)$$

MATHEMATICAL INVARIANT & GUARANTEE STRIP (EQUATIONS 1–22 ALIGNMENT)

1. Invariant Space $\mathbf{z}_{\text{inv}}$

$$\tilde{\Phi}_{\text{flow}} = \ln\left(1 + \frac{\sum A_{\text{out}}}{\sum A_{\text{in}} + \varepsilon}\right)$$

2. Harmonic Encoder Φ_{time}

$$w(\Delta t) = \sum_{b=1}^{3} \pi_b e^{-\lambda_b \Delta t} \cos(\omega_b \Delta t)$$

3. Edge-Trust Floor $\hat{g}_{ij}$

$$\hat{g}_{ij} \in [0.10, 1.0],\ \delta_{\text{floor}} = 0.10$$

4. Adaptive Gating & Head $\mathbf{h}_u^*$

$$\mathbf{h}_u^* = \alpha_u \mathbf{h}_u + (1 - \alpha_u)\mathbf{z}_{\text{inv}},\ \tau^* = 0.18$$